2. Study of different types of Network ca

Aim:-

Study of different types of Network Cable.

a) understand different type of network cable.

Different type of cables used in network are

1) unshielded Twisted pair (UTP) cable.

2) Shielded Twisted pair (STP) Cable

3) Coaxial cable.

4) Fibre optic Cable.

Cable type	Category	Maximum Data Transformation	Advantages Disadvantages	Application/ use	Image
UTP	Category 3	10 bps	Advantages * cheaper in cost * Easy to install as they have a smaller overall diameter. Disadvantage * More prone to (EM I). Electromagnetic interference and noise.	10 Base-T Ethernet Fast Ethernet Gigabit Ethernet	
	Category 5	up to 100 Mbps.		Fast Ethernet Gigabit Ethernet	
	category 5e.	1 Gbps.			

S/P.	category 6,6a	10Gbps.	Advantages * Shielded * Faster than UTP * Less Suspectible to noise and Interference	Gigabit Ethernet 10G Ethernet (55m) widely used in data centers.
SSTP	Category 7	10Gbps.	Disadvantages * Expensive * Gaeather installation Effort	Gigabit Ethernet 10G Ethernet (100m).
o-axial oaxial	RG-6 RG-59 RG-11	10-100Mbps.	* High bandwidth * versatile. Disadvantage * Limited distance	Speed of signal is 500m Television network high speed internet connection

fiber optics Cable	Single mode Multi-mode	100 Gbps.	Advantages * High Speed * High bandwidth * High Security * long distance Disadvantages. * Expensive * Require skilled installers	Maximum distance of fiber optics Cable is around 100 meters

b) Make your Ethernet Cross-over cable / Straight cable

*) Ethernet cabling CAT5e is certified for gigabit support, but CAT5 cabling works as well just over shorter distances

ii) optional two plug Shields

Step 1: To start construction of the device begin by threading Shielding onto the cable.

Step 2: Next, Strip approximately 1.5cm of cable Shielding from both ends. The crimping tool has a round area to complete this task

Step 3: After, you will need to untangle, the wires there should be four twisted pairs" Referring arrangement A and the other in B

Step 4: once the order is correct, bunch them together in a line and if there are any of you as shown.

Step 5: Next, push the cable right in, The match box

Result:

Therefore, the different types of Network cable are Successfully executed